

Combinatorics and Propositional Logic

Introduction:- Recursive Equations.

Master's Theorem is applicable to Recursive equation for Divide and Conquer approach.

$$T(n) = a(T(n/b)) + k$$

→ we get closed approximation.

But in 2nd type of equations.

$$\begin{cases} T(n) = T(n-1) + T(n-2) + k \\ T(1) = 1 \end{cases}$$

Imp

→ Sometimes you will get the answer just by putting values.

Sequence → Order of any object.

$$\begin{array}{ccccccc} \text{Ex} \rightarrow & 0, & 2, & 4, & 6, & 8, & 10, \dots \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ & n = & 0, & 1, & 2, & 3, & 4, & 5, \dots \end{array}$$

$$T(0) \rightarrow 0, T(1) \rightarrow 2, T(2) \rightarrow 4, \dots$$

Recurrence relation

$$T(n) = T(n-1) + 2 : n \geq 1$$

$$\text{Initial Cond}^n \rightarrow [T(0) = 0, T(1) = 2]$$

Solⁿ

Suppose solⁿ is $T(n) = 2n$ then it must satisfy the recurrence relation.

$$\text{LHS} = T(n) = 2n$$

$$\begin{aligned} \text{RHS} &= T(n-1) + 2 = 2(n-1) + 2 \\ &= 2n \end{aligned}$$

$$\underline{\text{LHS} = \text{RHS}}$$

So, $T(n) = 2n$ will be the solution.

Q → The recurrence eqn

$$T(1) = 1$$

$$T(n) = 2T(n-1) + n \quad n \geq 2$$

Evaluates to.

$$a) 2^{n+1} - n - 2$$

$$b) 2^n - n$$

$$c) 2^{n+1} - 2n - 2$$

$$d) 2^n + n$$

Put $n=1$ then found that only a and b are valid.
because $T(1)=1$.

$$\text{Then found } T(2) = 2T(2-1) + 2 \\ = 2T(1) + 2 \rightarrow 4$$

Only option a satisfies $T(2) \rightarrow 4$

So, Ans $\rightarrow$ (a).

$$\underline{\underline{Q}} \quad T(2^k) = 3T(2^{k-1}) + 1$$

$$T(1) = 1$$

$$\underline{\text{Sol}^n} \quad \text{put } 2^k = n$$

$$\text{then } T(n) = 3T(n/2) + 1$$

$$a=3, b=2 \quad k \geq 0$$

$$\underline{a < b^k \rightarrow O(n^{\log_2 3})}$$

$$T(n) \rightarrow O(n^{\log_2 3})$$

$$T(2^k) \rightarrow O(2^{k \log_2 3}) \rightarrow O(2^{\log_2 3^k}) = O(3^k)$$

Using Elimination method

$$a) T(2^k) = 2^k \quad \text{not} \quad \boxed{T(k) = 2^k}$$

**** Remember*

$$b) T(2^k) = (3^{k+1} - 1)/2$$

$$c) T(2^k) = 3^{\log_2 k}$$

$$d) T(2^k) = 2^{\log_3 k}$$

Option c and d are invalid because we get $T(1)$
at $k \geq 0$ because $T(2^0) = T(1)$

and in $3^{\log_2 k}$ and $2^{\log_3 k}$ at $k \geq 0$ log value
is not defined.

Q Let x_n denotes no. of binary string of length n contain no consecutive zero.
 which of the following recurrence does x_n satisfy.

- a) $x_n = 2x_{n-1}$ b) $x_n = x_{n/2} + 1$
 c) $x_n = x_{n/2} + n$ d) $x_n = x_{n-1} + x_{n-2}$

Solⁿ $x_1 = 2 \{01, 1\}$ $x_3 = 5 \{010, 011, 101, 110, 111\}$
 $x_2 = 3 \{01, 10, 11\}$

So the ans is $\rightarrow$ (d)

because $x_3 = x_{3/2} + 1$ in option b
 and $x_{3/2}$ can't be evaluated.

Generating functions

$\rightarrow$ Not useful in Computer science directly but useful in computation.

Series $\rightarrow a_1, a_2, a_3, a_4, a_5$

Task $\rightarrow$ Generate a generating fn of the series.

Approach $\rightarrow$ Give some weight to each term in series;

$$\underbrace{x^0 a_1 + a_2 x^1 + a_3 x^2 + a_4 x^3 + a_5 x^4}_{\downarrow \text{Generating function for series}}$$

$\rightarrow$ If sequence is not finite then generating fn will also infinite.

Ex $\rightarrow$ Sequence $\rightarrow 1, 1, 1, 1, \dots$

Generating fn $\rightarrow 1 + x + x^2 + x^3 + \dots$

Compact form of generating fn

1) $(x+1) = \frac{(x^2-1)}{(x-1)} = \frac{(1-x^2)}{(1-x)}$ 3) $1 + x + x^2 + \dots + x^n$
 $= \left(\frac{x^{n+1}-1}{x-1} \right) = \left(\frac{1-x^{n+1}}{1-x} \right)$

2) $(1+x+x^2) = \frac{(x^3-1)}{(x-1)} = \frac{(1-x^3)}{(1-x)}$

Pigeonhole Principle

If ' k ' is a positive integer and ' $k+1$ ' or more objects are placed into ' k ' boxes, then there is at least one box containing two or more objects.

Ex → If $k=3$ and $k+1=4$ then there are ^{many} ways to put the object in boxes such that atleast 1 box contain 2 or more objects.

Ex → Among any group of 367 people there must be atleast two same birthday.

Soln Max. days in a year can be possible → 366 (leap yr)
No. of people = 366+1

So allot each day to 366 people and allot any day to remaining 1.

In every case it is going to follow Pigeonhole principle.

Generalised Pigeonhole Principle

If ' N ' objects are placed in ' k ' boxes then atleast one box containing atleast $\lceil N/k \rceil$ objects. $[k < N]$

Propositional Calculus

Propo with logic

Proposition: Any declarative statement with truth values.

Certainty [My name is Aryan. → Proposition (T)
I am 12 yrs. old. → Proposition (false)

Uncertain [You are right or wrong → Not a proposition.

Propositional variable → Variable / symbol assign to a particular statement

Relation Ex → I am right: 'p'

Connectives → Things that connects more than one propositional statement

① $\sim p$: I am not right. [Negation]
 (or) $\sim p$: It is not the case that I am not right

② $\wedge$: p : I got 80% marks.
 q : I got A grade.] Conjunction
 $(p \wedge q)$: I got 80% marks and A grade

③ $\vee$: Disjunction
 $(p \vee q)$: I got 80% marks or A grade.

p	$\sim p$
T	F
F	T

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$ ^{XOR}
T	T	T	T	False
T	F	F	T	T
F	T	F	T	T
F	F	F	F	False

Implications

$p \longrightarrow q$
 (Hypothesis) (Conclusion)

p : You get 60 marks in Gate

q : You will go to IIT's

$p \rightarrow q$: If you get 60 marks in Gate then you will go to IIT's.

p	q	$p \Rightarrow q$
T	T	T
F	F	T
T	F	F
F	T	T

→ Implication have no relation with if-else in programming and general english.

→ If $(p \rightarrow q)$ is always true called 'Tautology'.

① If p then q

② If p, q

③ p is sufficient for q

④ q if p

⑤ q when p

→ a necessary condition for p is q.

→ q unless $\sim p$

→ p implies q

→ p only if q

→ A sufficient condition for q is p

- q whenever p
 - q is necessary for p
 - q follows from p
- These 3 and all previous are ways to represent $p \rightarrow q$.

for $(p \rightarrow q)$

Converse: $q \rightarrow p$

Contrapositive: $\sim q \rightarrow \sim p$

Inverse: $\sim p \rightarrow \sim q$

Bi-conditional: $p \leftrightarrow q : (p \rightarrow q) \wedge (q \rightarrow p)$

if and only if
(iff)

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

X-Nor



Consistent System: (3)

The system of statement is consistent if all the statements becomes 'True' simultaneously at a time.

Ex.

→ The diagnostic message is stored in buffer (p) it is transmitted. (q)

→ The diagnostic message is not stored in buffer ($\sim p$)

→ If the diagnostic message stored in buffer then it is transmitted. ($p \rightarrow q$) or ($\sim p \vee q$)

p	q	$\sim p$	$p \rightarrow q$	$p \vee q$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	F

All three are true if

$p \rightarrow \text{false}$
 $q \rightarrow \text{True}$

2nd method $\rightarrow$ To make all the statement true at a time do analysis like.

① $\sim p$ will true if $p \rightarrow \text{false}$.

② $(p \vee q)$ will true if $q \rightarrow \text{true}$ bcoz p is already false above

So $p \rightarrow \text{false}$, $q \rightarrow \text{true}$.

Equivalences $\rightarrow$ You can by-heart it, but don't need to do this because its easy to evaluate.

① $p \rightarrow q = \sim p \vee q = \sim q \rightarrow \sim p$

⑤ $Q \wedge \sim Q \rightarrow \phi$

② $p \vee q = \sim p \rightarrow q$

③ $p \wedge q = \sim (q \rightarrow \sim p)$

④ $\sim (p \rightarrow q) \equiv p \wedge \sim q$

Precedence Table

$\sim$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Priority decreases $\downarrow$

De-Morgan's Law

$\sim (p \vee q) \equiv \sim p \wedge \sim q$

$\sim (p \wedge q) \equiv \sim p \vee \sim q$

Ex $\rightarrow$ Ravi has a computer, and, a phone ($p \wedge q$)

$p \quad \wedge \quad q$

$\sim (p \wedge q) =$ Ravi doesn't have a computer
or
Ravi doesn't have a phone

Argument / Inference: If a set of premises $\{p_1, p_2, \dots, p_n\}$ yield another proposition Q (conclusion) the whole process is called argument.

Note: An argument is said to be valid if the conclusion Q can be derived from the premises by applying the rule of inferences.

An argument is denoted by $\{p_1, p_2, p_3, \dots, p_n\} \vdash Q$

Proposition which are 'true' called as Premises

Rules of Inference

Representation of a valid argument.

1. The argument $\{p_1, p_2, \dots, p_n\} \vdash Q$ is valid.
2. The statement $\{p_1, p_2, \dots, p_n\} \Rightarrow Q$ is true.
3. The statement formula

$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow Q$ is a tautology.

1. Simplification Rule

①) $\frac{P \wedge Q \text{ valid}}{P/Q}$ or $(P \wedge Q) \Rightarrow P$ is true / Q is true
 $(P \wedge Q) \rightarrow P$ is tautology / Q is tautology

2) Addition Rule

a) $\frac{P}{P \vee Q}$ valid if $P \rightarrow \text{true}$ then $(P \vee Q) \rightarrow \text{true}$

b) $\frac{Q}{P \vee Q}$ same as above $Q \rightarrow \text{true}$ then $(P \vee Q) \rightarrow \text{true}$

or $P \rightarrow (P \vee Q)$ is true $Q \rightarrow (P \vee Q)$ is true

③ $\frac{\sim P}{P \rightarrow Q}$ valid

④ $\frac{Q}{P \rightarrow Q}$: valid $[P \rightarrow Q = \sim P \vee Q]$

⑤ $\frac{\sim(P \rightarrow Q)}{P}$ valid

⑥ $\frac{\sim(P \rightarrow Q)}{\sim Q}$ valid

$\sim(P \rightarrow Q) = P \wedge \sim Q$ is true if

P is true and $\sim Q$ is true.

7. Conjunction $\frac{Q}{P} \text{ valid}$
 $\frac{P}{P \wedge Q}$

8. Disjunctive Syllogism $\frac{P \vee Q}{\sim P}$
 $\frac{}{Q} \text{ valid}$

9. Conjunctive syllogism $\frac{\sim(P \wedge Q)}{P}$
 $\frac{}{\sim Q} \text{ valid}$

10. Modus Ponens / Rule of detachment $\frac{P \rightarrow Q}{P}$
 $\frac{}{Q} \text{ valid}$

11. Modus Tollens or Rule of contrapositive $\frac{P \rightarrow Q}{\sim Q}$
 $\frac{}{\sim P} \text{ valid}$

12. Transitivity $\frac{P \rightarrow Q}{Q \rightarrow R}$
 $\frac{}{P \rightarrow R} \text{ valid}$

13. Dilemma $\frac{P \vee Q}{P \rightarrow R}$
 $\frac{Q \rightarrow R}{R} \text{ valid}$

14. Constructive Dilemma $\frac{P \vee Q}{P \rightarrow R}$
 $\frac{Q \rightarrow S}{R \vee S} \text{ valid}$

15. Destructive Dilemma $\frac{P \rightarrow R}{Q \rightarrow S}$
 $\frac{\sim R \vee \sim S}{\sim P \vee \sim Q} \text{ valid}$

Invalid arguments (fallacies)

① fallacy of assuming converse

$\frac{P \rightarrow Q}{Q}$
 $\frac{}{P} \text{ Not Valid}$

$\sim P \vee Q$ if Q is true then P can be T/F any thing but here it considers P as true only so, fallacy.

But if you assume the converse as

$$\frac{Q \rightarrow P}{Q} \quad \underline{P}$$

then it is valid.

② fallacy of assuming inverse

$$\frac{P \rightarrow Q}{\sim P} \quad \underline{\sim Q}$$

③ Non Sequitur

$$\frac{P}{Q} \quad \text{Not valid}$$

~~5~~

Here p, q are not related.

Q Which of following are not valid?

a) The conclusion $\sim p$ follows from premises $\{P \rightarrow Q, Q \rightarrow R, \sim R\}$

$$\frac{P \rightarrow Q}{Q \rightarrow R} \quad \underline{P \rightarrow R} \quad (\text{transitivity})$$

$$\frac{\sim R}{\sim P} \quad | \text{Modus tollens}$$

so valid

b) $(p \rightarrow q)$ follows $\{ \sim p, p \vee q \}$

$$\frac{\sim p}{p \vee q} \quad \underline{q}$$

then $q \vee \sim p$ also true
($p \rightarrow q$)

$A \vee B$

$B \rightarrow C$

$A \rightarrow D$

$\sim D$

C

$A \vee B$

B

$(\sim A \vee D) \wedge \sim D \Rightarrow \sim A$

$B \wedge (\sim B \vee C) = C$

g) Consider the following set of premises which of the following is true

$$S_1: A \rightarrow B, A \rightarrow C, B \rightarrow \sim C, A$$

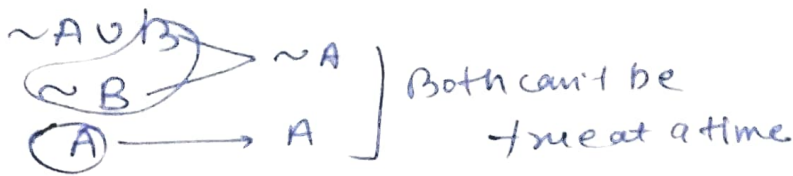
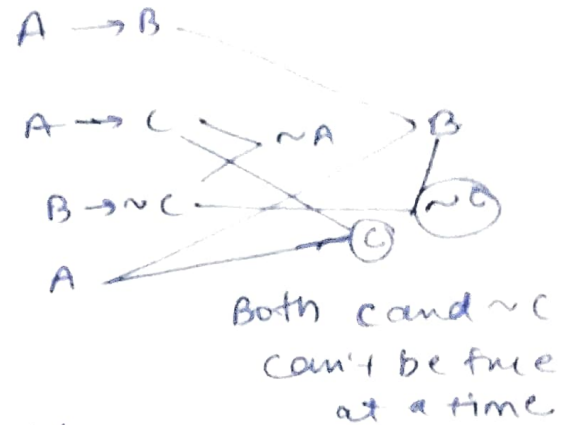
$$S_2: \sim A \vee B, \sim B, A$$

a) S_1 is inconsistent

b) S_2 _____

c) both S_1 and S_2 are inconsistent

d) _____ are consistent



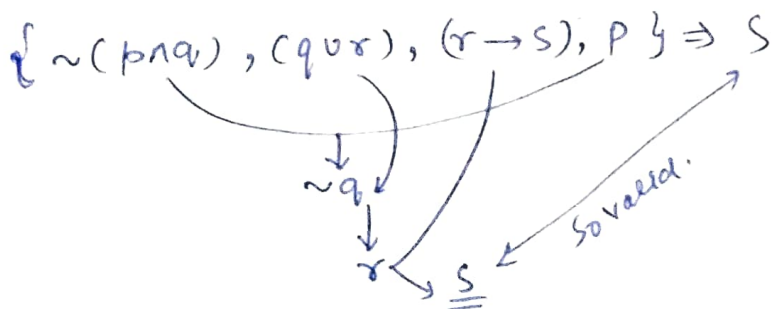
Conditional proof (6)

If a set of premises $\{P_1, P_2, P_3 \dots P_n\} \Rightarrow R$ then

$$\{P_1, P_2 \dots P_n\} \Rightarrow (A \rightarrow R)$$

Ex $\rightarrow$ Check whether the argument is valid or not.

$$\{\sim(P \wedge Q), (Q \vee R), (R \rightarrow S)\} \Rightarrow (P \rightarrow S)$$



Proposition function

Proposition statement: $[2 \text{ is greater than } 10] \text{ false}$

function: $P(x) = [x \text{ is greater than } 10] \text{ True/false}$

Variable Predicate

Ex $P(2) \rightarrow \text{false}$

$P(12) \rightarrow \text{True}$

② $R(x)$ = Ravi has a x phone

$R(\text{samsung}), R(\text{Apple})$ —

Computer 'x' is connected to network 'y' $\rightarrow P(x, y)$

Quantifiers: [Do atleast 20 question on this topic]

$P(x)$: $x > 2$

Domain of Disclosure: $\{1, 2, 3, 4\}$ [Value that x can take]

$P(1) \rightarrow F$ $P(2) \rightarrow F$ $P(3) \rightarrow T$ $P(4) \rightarrow T$

① $\forall x P(x)$ $\rightarrow$ If the proposition function $P(x)$ is true for all x then it returns True.
for each x
for all x
for every x
So, for $P(x)$ $x > 2$
 $\forall x P(x) \rightarrow \text{False}$

If Domain = $\{3, 4\}$
then $P(x) = x > 2$ is following $\forall x P(x)$

Here, $\forall x$ $\rightarrow$ quantifier.

② $\exists x P(x)$ $\rightarrow$ If atleast one value in domain for which $P(x)$ is True.

True exist
Some

$$\begin{aligned}\forall x P(x) &= P(3) \cap P(4) \\ &= T \cap T \\ &= T\end{aligned}$$

Union $\leftarrow$

$$\begin{aligned}\exists x P(x) &= P(3) \cup P(4) \\ &= T \cup T \\ &= T\end{aligned}$$

Examples

$P(x)$: $x > 3$

Domain: $\{4, 5, 6\}$

$$\begin{aligned}\forall x P(x) &\rightarrow P(4) \cap P(5) \cap P(6) \\ &\rightarrow T \cap T \cap T \\ &\rightarrow T\end{aligned}$$

$$\begin{aligned}\exists x P(x) &\rightarrow P(4) \cup P(5) \cup P(6) \\ &\rightarrow \underline{\text{True}}\end{aligned}$$

Imp

$\Rightarrow$ If $\forall x P(x)$ is true for same domain and propositional function $\exists x P(x)$ will always True, but its converse is not always True.

③

⇒ $\forall$ and $\exists$ have high precedence over other connectives

Ex. $\forall x P(x) \vee Q(x) \rightarrow ((\forall x P(x)) \vee Q(x))$

Result
Result

Free and Bounded Variable

$\forall x P(x) \vee Q(x)$

↓
Bounded with quantifier.

↪ free

Scope

$\forall x P(x) \vee Q(x)$

↪
($\forall x$) have scope upto $P(x)$ only

Distribution Quantifiers

$\forall x (x+y+z \neq 10) \cong (\forall x (x)) + y + z = 10$

↪ Only x is bounded here

$\exists x P(x) \vee \forall x Q(x) \cong \exists x P(x) \vee \forall y Q(y)$

↪ Similar to LHS but easy to understand.

x is bounded to $\exists$ x is bounded to $\forall$

Ex: $\forall x P(x) = P(x_1) \wedge P(x_2) \wedge P(x_3)$

$D: \{x_1, x_2, x_3\}$

$\forall x (P(x) \wedge Q(x)) \Rightarrow \frac{(P(x_1) \wedge Q(x_1)) \wedge (P(x_2) \wedge Q(x_2))}{\text{same}}$

$D: \{x_1, x_2\}$

$\forall x P(x) \wedge \forall x Q(x) = P(x_1) \wedge P(x_2) \wedge Q(x_1) \wedge Q(x_2)$

$D: \{x_1, x_2\}$

(a) So, $\forall$ quantifier is distributive over $\wedge$

$\forall x (P(x) \vee Q(x)) = (P(x_1) \vee Q(x_1)) \wedge (P(x_2) \vee Q(x_2))$

$\forall x P(x) \vee \forall x Q(x) = (P(x_1) \wedge P(x_2)) \vee (Q(x_1) \wedge Q(x_2))$

So $\forall$ is not distributive over $\vee$.

x (3) VISM

	Conjunction ($\wedge$)	Disjunction ($\vee$)
$\forall$	Distributed	Non-Distributed
$\exists$	Non-distributed Distributed	Distributed

Quantifiers with negation

$$\sim(\forall x P(x))$$

Ex: Every student in class has taken a

$$\sim(\exists x P(x))$$

Course in CS.

or

$P(x)$: x has taken a course in CS.

$$\forall x P(x)$$

Domain: student of class.

$\sim(\forall x P(x)) \rightarrow$ No student in class has taken a course in CS.

$$\boxed{\sim \forall x P(x) \equiv \exists x \sim P(x)}$$

There exist at least one student who has not taken a course in CS.

$$\boxed{\sim \exists x P(x) \equiv \forall x \sim P(x)}$$

Important examples

$$\begin{aligned} \sim \forall x \frac{(P(x) \wedge Q(x))}{P(x)} &\equiv \exists x \sim P(x) \\ &\equiv \exists x (\sim P(x) \vee \sim Q(x)) \end{aligned}$$

$$\sim \forall x \frac{(P(x) \rightarrow Q(x))}{(\sim P(x) \vee Q(x))} \equiv \exists x (P(x) \wedge \sim Q(x))$$

$$\begin{aligned} \sim \exists x (P(x) \wedge Q(x)) &\equiv \forall x (\sim P(x) \vee \sim Q(x)) \\ &\equiv \forall x (P(x) \rightarrow \sim Q(x)) \end{aligned}$$

$$\sim \exists x (P(x) \rightarrow Q(x)) \equiv \forall x (P(x) \wedge \sim Q(x))$$

Ex: Not all that glitter is gold

a) $\forall x: \text{glitter}(x) \rightarrow \sim \text{gold}(x)$

b) $\forall x: \text{gold}(x) \wedge \sim \text{glitter}(x)$

c) $\exists x: \text{gold}(x) \wedge \sim \text{glitter}(x)$

d) $\exists x: \text{glitter}(x) \wedge \sim \text{gold}(x)$

$p(x) \Rightarrow \forall x: \text{glitter} \rightarrow \text{gold}(x) \quad (P \rightarrow Q)$

$\sim p(x): \exists x: \text{glitter}(x) \wedge \sim \text{gold}(x).$

using $\sim \forall x (P \rightarrow Q) = \exists x (P \wedge \sim Q)$

9 important points to note:-

$\Rightarrow \forall x (P(x) \Rightarrow Q(x)) \Rightarrow (\forall x (P(x) \Rightarrow Q(x)))$

✓ Whenever a predicate is satisfiable then its negation is also satisfiable.

→ In Substrings select the continuous elements.

→ No. of ways of distributing 'n' similar things among r different things equals to $(n+r-1)C_r$

→ Ex: Distributing 10 flowers among 2 girls = $11C_{10} = 11$

→ No. of derangement (Q. 3.16 : Combinatorics).

✓ Tautology (Always 1), Contradiction (Always 0),
Contingency (0/1).

✓ $p \equiv q$ denotes p, q are logically equivalent

$\Rightarrow \forall x (P(x) \wedge Q(x)) \leftrightarrow \forall x P(x) \wedge \forall x Q(x)$

$(\forall x P(x) \vee \forall x Q(x)) \rightarrow \forall x (P(x) \vee Q(x))$

$\exists x (P(x) \vee Q(x)) \leftrightarrow \exists x P(x) \vee \exists x Q(x)$

$\exists x (P(x) \wedge Q(x)) \rightarrow \exists x P(x) \wedge \exists x Q(x)$

Day 3

Discrete Maths (Part-1: P2)

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① $p \rightarrow q$ can be write as: p is sufficient for q , q is necessary for p , p only if q , q unless $\neg p$, p implies q , q if p , q when p , q if p, q etc

② $p \leftrightarrow q$ is equivalent to X-NOR, and $(p \rightarrow q) \wedge (q \rightarrow p)$.
 ③ System is consistent if all the logics of system is true
 ④ $p \rightarrow q \equiv \sim p \vee q$.

⑤ Inference Rules

① Disjunctive Syllogism:
$$\frac{P \vee Q}{\sim P} \quad Q$$
 ② Conjunctive Syllogism:
$$\frac{\sim (P \wedge Q)}{P} \quad \sim Q$$

③ Modus Ponens / Rule of detachment:
$$\frac{P \rightarrow Q}{P} \quad Q$$
 ④ Modus Tollens / Rule of contrapositive:
$$\frac{P \rightarrow Q}{\sim Q} \quad \sim P$$

⑤ Transitivity:
$$\frac{P \rightarrow Q}{Q \rightarrow R} \quad P \rightarrow R$$
 ⑥ Dilemma:
$$\frac{P \vee Q}{P \rightarrow R} \quad Q \rightarrow R \quad R$$

⑦ $[\{ p_1, p_2, p_3 \dots p_n \} \wedge Q] \Rightarrow R$ can be written as $\{ p_1, p_2, p_3 \dots p_n \} \Rightarrow (Q \rightarrow R)$.

⑧ Quantifiers: $\forall (x) P(x) = P(x_1) \wedge P(x_2) \wedge \dots$
 $\exists x P(x) = P(x_1) \vee P(x_2) \vee \dots$

eg $\forall x P(x)$ is true then in same domain $\exists x P(x)$ is true but converse is not true

	$\wedge$	$\vee$	
$\forall (x)$	Distributed	Non distributed	$\sim \forall (x) P(x) \equiv \exists (x) \sim P(x)$
$\exists (x)$	Non-Distributed	Distributed	$\exists (x) \sim P(x)$

$\rightarrow p \equiv q$ denotes p and q are logical equivalents

Imp Question: $\sim \forall (x) P(x) \vee \sim \forall (x) Q(x) \rightarrow \forall (x) [P(x) \vee Q(x)]$
 $\downarrow$
not both side.

$$\rightarrow \neg \forall x [P \Rightarrow Q] \Rightarrow [\forall x P(x) \Rightarrow \neg \exists x Q(x)]$$

$\rightarrow$ Tautology $\rightarrow$ All true

Contingency $\rightarrow$ True / false mixed

Contradiction $\rightarrow$ All false

$\rightarrow$ Satisfiable : Either tautology or contingency.

Unsatisfiable : Contradiction.

$\rightarrow$ If predicate logic is satisfiable then its negation is also satisfiable

$\rightarrow$ In CNF, for any formula, there is a truth assignment for which atleast half the clause evaluates to true.

$$\rightarrow \exists x [(P(x) \vee Q(x)) \leftrightarrow \exists x (P(x) \vee Q(x))$$

and

$$\exists x (P(x) \wedge Q(x)) \leftrightarrow \exists x P(x) \vee \exists x Q(x)$$

Both side distributed

$\rightarrow$ In case of every use $\forall x A(x) \rightarrow$, and with some use $\exists x A(x) \wedge$.

$$\rightarrow \exists x (A(x) \wedge B(x)) \rightarrow \exists x A(x) \wedge \exists x B(x)$$

$\rightarrow \exists! x$ means only or exactly 1.

$\rightarrow$ All the sets over rational no. and integers are countable